

Hierarchical Cluster Analysis for Customer Segmentation

```
dpath = 'customerSegmentationLab.xlsx'; % dataset path
T = readtable(dpath, "Sheet", "Foglio1", 'ReadVariableNames',true,
'ReadRowNames',true);
```

```
customers = T.Properties.RowNames; % names of customers
features = T.Properties.VariableNames; % features of customers
```

```
head(T); % first few rows of data
```

	Age	AnnualIncome	SpendingScore	OnlineVisits	AvgPurchaseValue	DiscountSensitivity
Customer 1	23	28	82	24	38	79
Customer 2	25	32	77	21	42	74
Customer 3	22	26	88	26	35	81
Customer 4	27	35	80	19	46	69
Customer 5	24	30	85	22	40	77
Customer 6	26	34	79	20	44	72
Customer 7	45	75	62	11	95	35
Customer 8	48	82	58	10	102	29

```
X = table2array(T); % numerical matrix of T
[n, p] = size(X);
```

```
% rows represent customers and customers' in columns
fprintf('Rows = %d, Columns = %d', [n, p]);
```

```
Rows = 30, Columns = 6
```

```
Z = zscore(X); % robust standardization
```

```
% euclidean distance or dissimilarity matrix
d = pdist(Z, 'euclidean'); % pairwise Euclidean distances
D = squareform(d); % square distance matrix
```

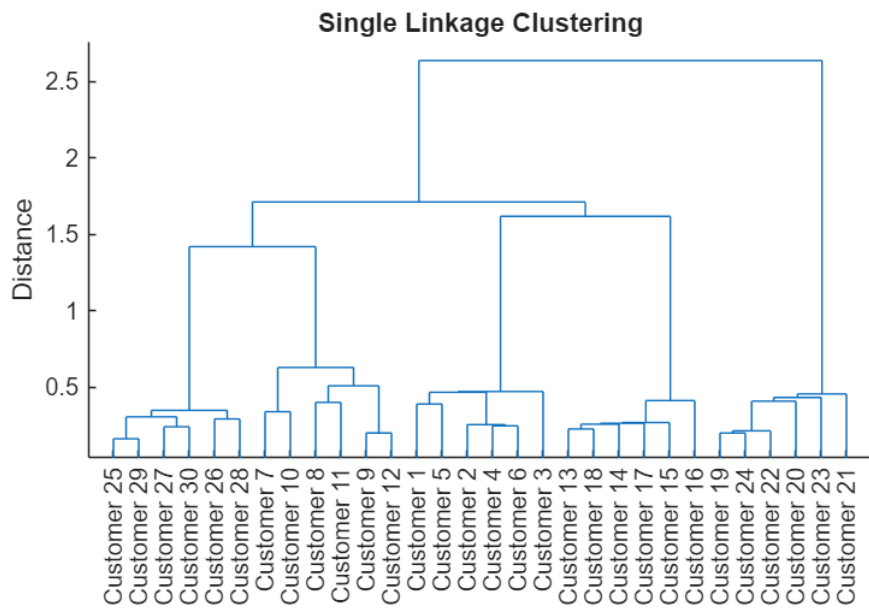
Linkage Methods and Dendrograms

Hierarchical Clustering using four linkage methods

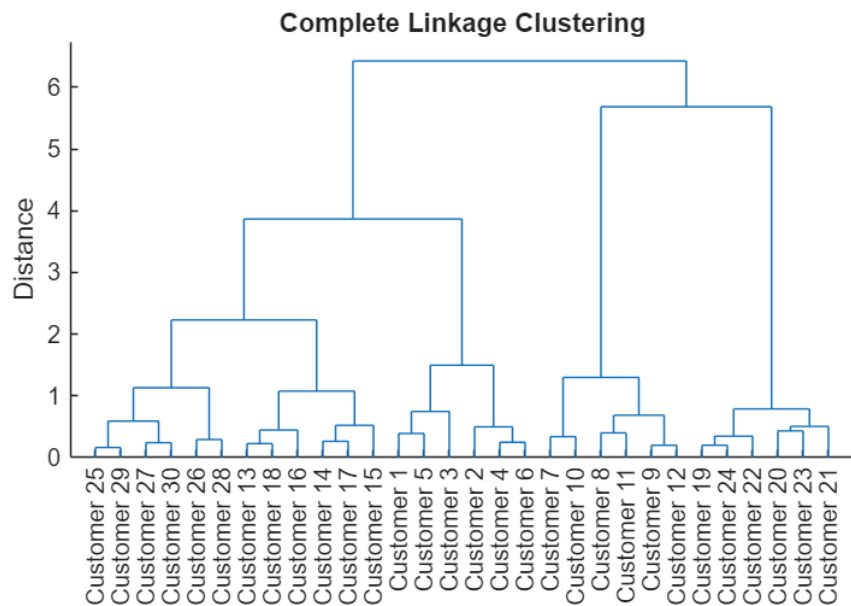
```
% linkage structures
LSig = linkage(D, "single");
LCom = linkage(D, "complete");
LAvg = linkage(D, 'average');
LWard = linkage(D, 'ward');
```

Plotting Dendrograms

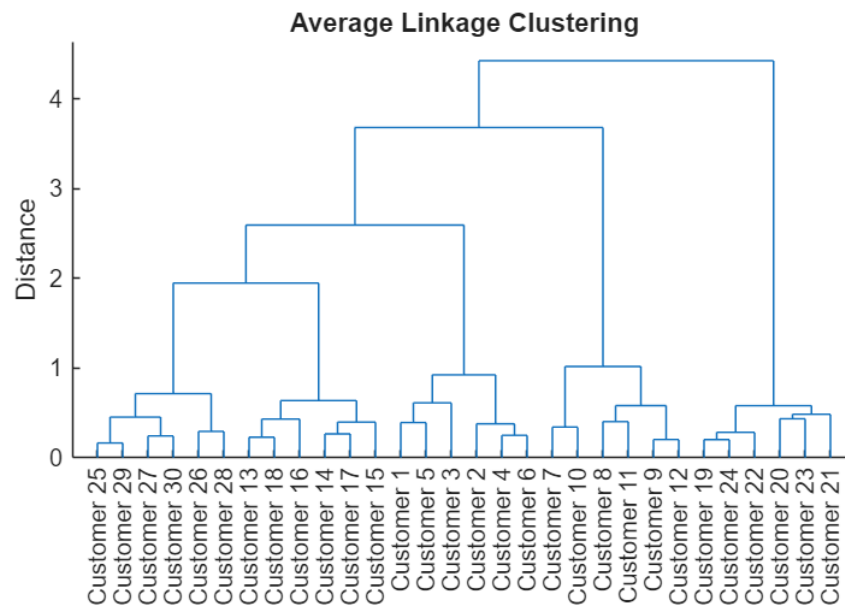
```
% Create dendrograms to visualize the hierarchical clustering
figure('Name', 'Single Linkage');
dendrogram(LSig, 0, "Labels", customers);
title('Single Linkage Clustering');
ylabel('Distance');
```



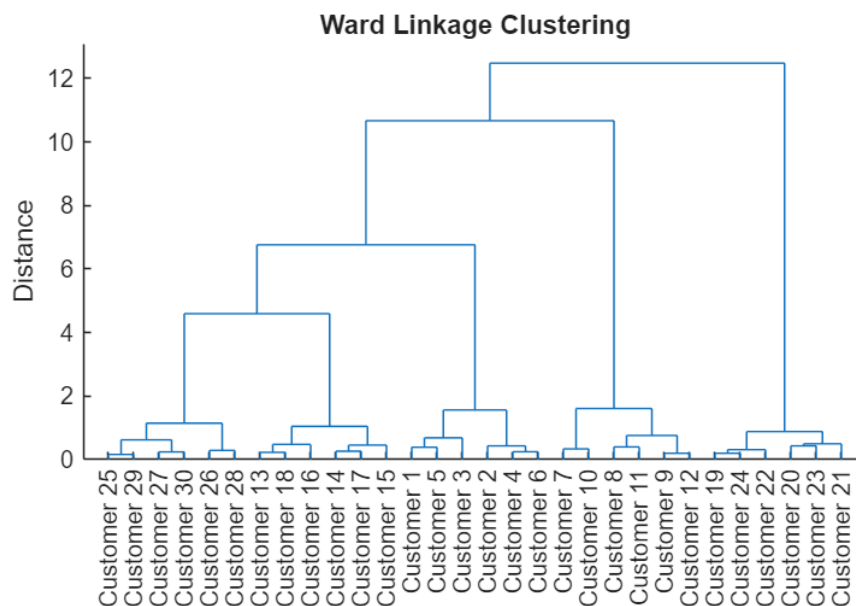
```
figure('Name', 'Complete Linkage');
dendrogram(LCom, 0, "Labels", customers);
title('Complete Linkage Clustering');
ylabel('Distance');
```



```
figure('Name', 'Average Linkage');
dendrogram(LAvg, 0, "Labels", customers);
title('Average Linkage Clustering');
ylabel('Distance');
```



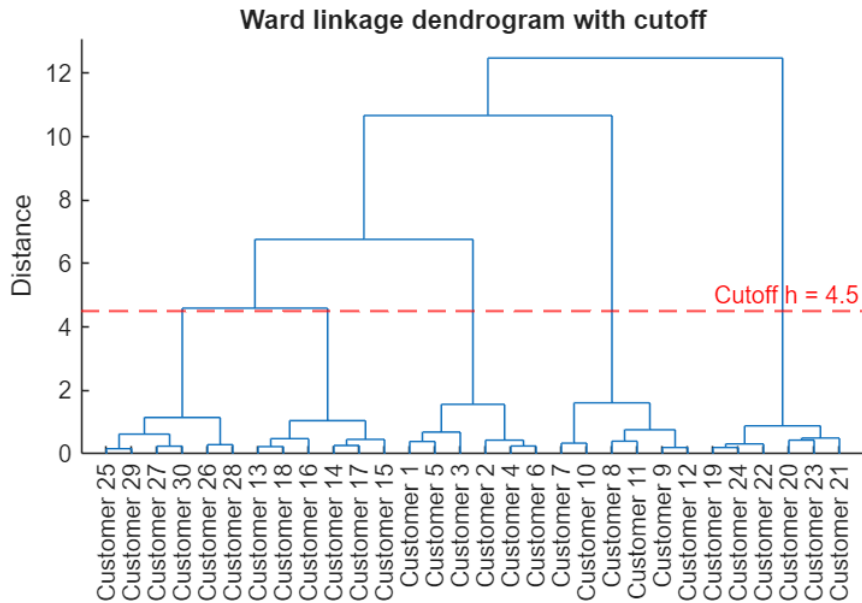
```
figure('Name', 'Ward Linkage');
dendrogram(LWard, 0, "Labels", customers);
title('Ward Linkage Clustering');
ylabel('Distance');
```



Ward Linkage and Hierarchical Clustering

```
cuttOff = 4.5; % cutt-off value (distance threshold)

figure('Name','Ward dendrogram with cutoff');
dendrogram(LWard, 0, "Labels", customers);
title("Ward linkage dendrogram with cutoff");
ylabel("Distance");
yline(cuttOff, 'r--', sprintf('Cutoff h = %.1f', cuttOff), 'LineWidth', 1);
```



Extract clusters using the cut-off height

```
clus = cluster(LWard, "criterion", "distance", "cutoff", cuttOff); % make partitions
with a cutt-off
c5Ward = cluster(LWard, "maxclust", 5);

% scatter plot of customers' Annual Income and Spending Score
gscatter(T.AnnualIncome, T.SpendingScore, c5Ward);
xlabel('Annual Income');
ylabel('Spending Score');
title('Customers grouped by Ward clustering (3 clusters)');
```

